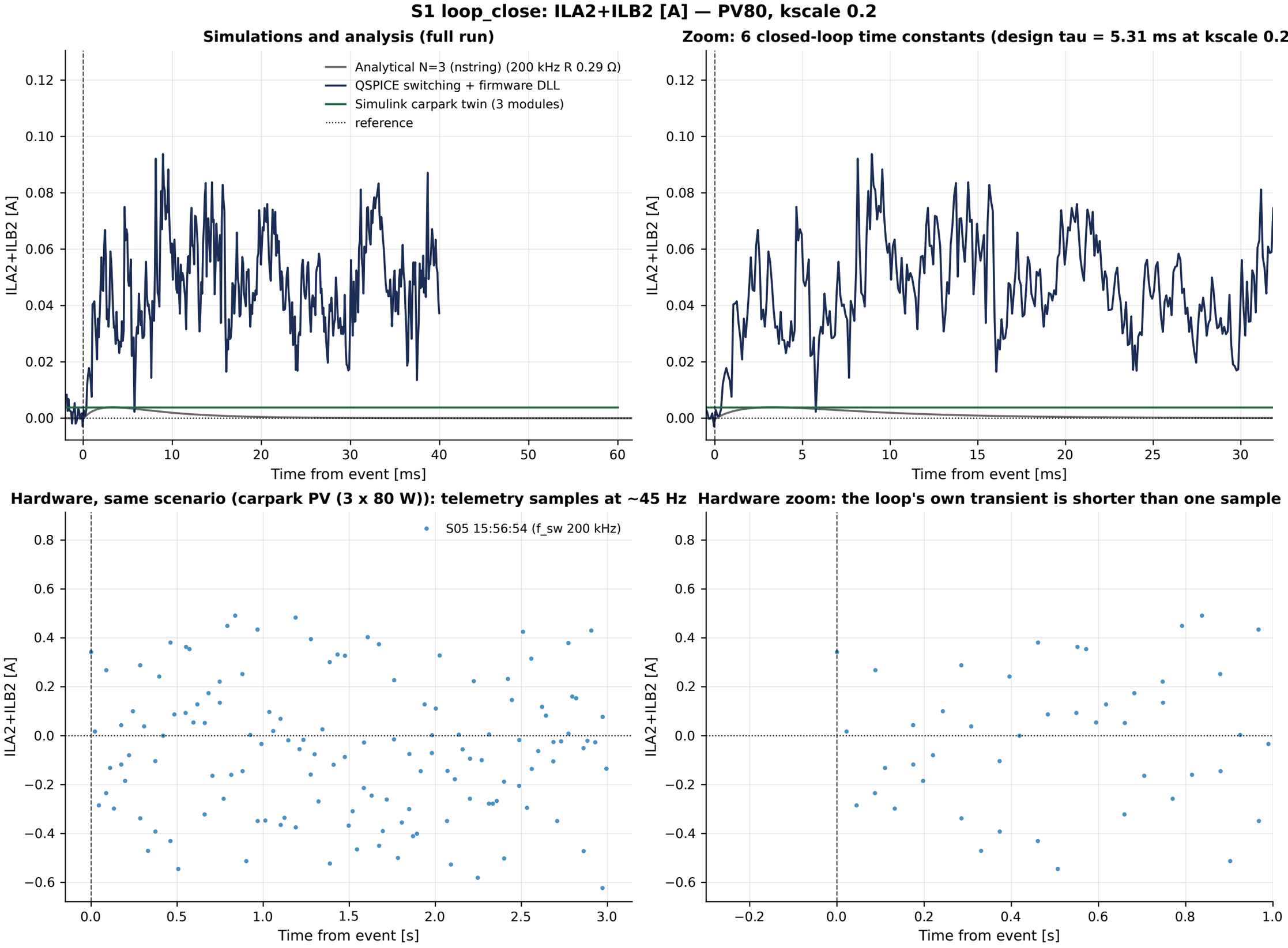
S1 loop_close: ILA2+ILB2 [A] — PV80, kscale 0.2
Simulations and analysis (full run)
Zoom: 6 closed-loop time constants (design tau = 5.31 ms at kscale 0.2
Analytical N=3 (nstring) (200 kHz R 0.29 Ω)
QSPICE switching + firmware DLL
Simulink carpark twin (3 modules)
reference
ILA2+ILB2 [A]
Time from event [ms]
Hardware, same scenario (carpark PV (3 x 80 W)): telemetry samples at ~45 Hz
Hardware zoom: the loop's own transient is shorter than one sample
S05 15:56:54 (f_sw 200 kHz)
Time from event [s]